

A horizontal banner with a dark green background. It features a faint, semi-transparent image of an Excel spreadsheet. The spreadsheet contains various data points, including dates (e.g., 11/2/2014, 11/9/2014, 11/16/2014), numerical values, and percentages. There are also small bar charts and line graphs visible within the spreadsheet grid.

LOOKUP & REFERENCE FUNCTIONS

Using **Named Arrays** can simplify a lookup function if you use the same data array in multiple formulas

For example, if you name the array from A1:D6 "Apparel"...

	Apparel			Product
	A	B	C	D
1	Product	Quantity	Product ID	Price
2	T-shirt	26	93754	\$14.99
3	Sweater	14	24783	\$49.99
4	Shorts	22	23984	\$24.50
5	Socks	36	58394	\$9.99
6	Spandex Unitard	2	27838	\$79.99

...you can write your vlookup formula in either of the following ways:

=VLOOKUP(A1,\$A\$1:\$D\$6,2)

=VLOOKUP(A1,Apparel,2)

Let's take a look at one of Excel's most common reference functions – **VLOOKUP**:

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

This is the **value** that you are trying to match in the table array

This is **where** you are looking for the lookup value

Which **column** contains the data you're looking for?

Are you trying to match the **exact** lookup value (0), or something similar (1)?

	A	B	C	D
1	Product	Quantity	Product ID	Price
2	T-shirt	26	93754	\$14.99
3	Sweater	14	24783	\$49.99
4	Shorts	22	23984	\$24.50
5	Socks	36	58394	\$9.99
6	Spandex Unitard	2	27838	\$79.99

D2=VLOOKUP(A2, \$G\$1:\$H\$5, 2, 0)

G	H
Product	Price
Shorts	\$24.50
Sweater	\$49.99
Spandex Unitard	\$79.99
T-shirt	\$14.99
Socks	\$9.99

To populate the Price in column D, we look up the name of the product in the data array from G1:H5 and return the value from the 2nd column over

Use **HLOOKUP** if your table array is transposed (variables headers listed in rows)

=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

This is the **value** that you are trying to match in the table array

This is **where** you are looking for the lookup value

Which row contains the data you're looking for?

Are you trying to match the **exact** lookup value (0), or something similar (1)?

	A	B	C	D
1	Product	Quantity	Product ID	Price
2	T-shirt	26	93754	\$14.99
3	Sweater	14	24783	\$49.99
4	Shorts	22	23984	\$24.50
5	Socks	36	58394	\$9.99
6	Spandex Unitard	2	27838	\$79.99

D2=HLOOKUP(A2, \$H\$1:\$L\$2, 2, 0)

With an HLOOKUP, we search for the product name in F1:J2 and return the value from the 2nd row down

G	H	I	J	K	L
Product	Shorts	T-shirt	Sweater	Spandex Unitard	Socks
Price	\$24.50	\$14.99	\$49.99	\$79.99	\$9.99

There are **two key rules** that constrain **VLOOKUP** and **HLOOKUP** formulas:

1. The lookup value must be in the **first column** of a VLOOKUP table array or the **first row** of a HLOOKUP table array
2. Excel will always return the value from the **top most row** or **left most column** of a table array when multiple instances of the lookup value are present



PRO TIP:



Avoid breaking Law #2 by identifying a “Key” that is common to both datasets and is unique for every row (NOTE: Keys often take the form of a concatenation of multiple fields)

The **ROW** function returns the row number of a given *reference*, while the **ROWS** function returns the number of rows in a given *array* or *array formula*

=ROW([reference])

=ROWS(array)

This example uses an array, which is why it includes the fancy {} signs – more on that in the ARRAY functions section

ROW(C10) = 10

ROWS(A10:D15) = 6

ROWS({1,2,3;4,5,6}) = 2

The **COLUMN** function returns the column number of a given *reference*, while the **COLUMNS** function returns the number of columns in a given *array* or *array formula*

=COLUMN([reference])

=COLUMNS(array)



PRO TIP:

Leave the cell reference out and just write ROW() or COLUMN() to return the row or column number of the cell in which the formula is written

COLUMN(C10) = 3

COLUMNS(A10:D15) = 4

COLUMNS({1,2,3;4,5,6}) = 3

The **INDEX** function returns the *value* of a specific cell within an array

=INDEX(array, row_num, column_num)

What range of cells
are you looking at?

How many rows
down is the value
you want?

How many columns
over is the value you
want?

	A	B	C
1	Tools	Price	Inventory
2	Hammer	\$5.00	55
3	Screw Driver	\$2.50	66
4	Pliers	\$3.34	333
5	Wrench set	\$10.00	234
6	Chain Saw	\$55.48	23
7	Tool Box	\$19.99	5
8	Level	\$2.25	7

INDEX(\$A\$1:\$C\$5, 5, 3) = 234

*In this case we're telling Excel to find the value of a cell somewhere within the array of A1:C5. Starting from the upper left, we move down to the **5th row** and right to the **3rd column**, to return the value of **234***

The **MATCH** function returns the *position* of a specific value within a column or row

=MATCH(lookup_value, lookup_array, [match_type])

What value are you trying to find the position of?

In which row or column are you looking? (**must be a 1-dimensional array**)

Are you looking for the exact value (0), or anything close?

1: Find largest value $\leq$ or = lookup_value

0: Find exact lookup_value

-1: Find smallest value $\geq$ or = lookup_value

	A	B
1	Tools	Price
2	Hammer	\$5.00
3	Screw Driver	\$2.50
4	Pliers	\$3.34
5	Wrench set	\$10.00

MATCH("Pliers",\$A\$1:\$A\$5, 0) = 4

	A	B	C
1	Tools	Price	Inventory
2	Hammer	\$5.00	55
3	Screw Driver	\$2.50	66
4	Pliers	\$3.34	333

MATCH(66,\$A\$3:\$C\$3, 0) = 3

*Matching the word "Pliers" in column A, we find it in the **4th** row. Matching the number 66 in row 3, we find it in the **3rd** column*

INDEX and **MATCH** are commonly used in tandem to act like a LOOKUP function; the only difference is that **INDEX/MATCH** can find values in any column or row in an array

Example: Price Checker

	A	B	C	D
1		Small	Medium	Large
2	Sweater	\$10	\$12	\$15
3	Jacket	\$30	\$35	\$40
4	Pants	\$25	\$30	\$35
5				
6	Product:	Pants		
8	Size:	Medium		
10	PRICE:	?		
11				

In this example, we want to populate the price of a given product and size in cell B10 by returning a particular value within the array B2:D4

B10=INDEX(B2:D4, MATCH(B6,A2:A4,0), MATCH(B8,B1:D1,0))

The number of rows down to index depends on what product I'm looking for, so we use a MATCH function and search for the value in cell B6 (in this case "Pants")

The number of columns over to index depends on what size I'm looking for, so we use a MATCH function and search for the value in cell B8 (in this case, "Medium")

Considering the output of each MATCH function, the formula is just a simple INDEX:

B10 = INDEX(B2:D4, 3, 2) = \$30

XLOOKUP can retrieve values from a table or range by matching a lookup value, and offers more flexibility than **VLOOKUP**, **HLOOKUP**, or **INDEX & MATCH** formulas

=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])



Which **value** are you looking to match?



Where are you trying to find a match for your lookup value?



Where are the **values** you want to retrieve?



What if the lookup value **isn't found** in the lookup array?



Are you looking for an **exact, approximate, or wildcard** match?



Do you want to search **top down** or **bottom up**?

IMPORTANT NOTE: XLOOKUP is currently only available for **Office 365** subscribers

XLOOKUP

- ✓ Can retrieve a **dynamic array** of results
- ✓ Can lookup values **anywhere** in an array (left or right, horizontal or vertical)
- ✓ Defaults to **exact match**
- ✓ Supports native **wildcard** text matching
- ✓ Includes **built-in error handling** when a lookup value is not found
- ✓ Can find approximate matches in **unsorted** lists
- ✓ Can search **top-down** or **bottom-up**

VLOOKUP

- ✗ Can only return a **single value**
- ✗ Can only lookup values to the **right**, requires **HLOOKUP** for horizontal matching
- ✗ Defaults to **approximate match**
- ✗ Does not natively support **wildcard** matching
- ✗ Requires an additional **IFERROR** function for error handling
- ✗ Requires **sorted** lists for approximate matching
- ✗ Only searches **top-down**

The **CHOOSE** function selects a value, cell reference, or function to perform from a list, based on a given index number

=CHOOSE(index_num, value1, [value2], ...)



Which item in the following list should be evaluated?

1st item in the list

2nd item in the list

3rd, 4th, 5th, etc...

FUN FACTS ABOUT CHOOSE:

- List items can include **numbers, cell references, defined names, formulas, or text** (*or a mix!*)
- CHOOSE acts like an **INDIRECT** function, and can interpret cell references instead of treating them as text
- You can combine CHOOSE with other functions, or nest it directly into a cell reference

The **OFFSET** function is similar to **INDEX**, but can return either the value of a cell within an array (like INDEX) or a specific *range* of cells

=OFFSET(reference, rows, columns, [height], [width])

What's your
starting
point?

How many
rows down
should you
move?

How many
columns over
should you
move?

If you want to return a
multidimensional array,
how tall and wide should it
be?

An **OFFSET** formula where [height]=1 and [width]=1 will operate exactly like an INDEX. A more common use of **OFFSET** is to create dynamic arrays (like the Scroll Chart example in the appendix)



PRO TIP:

Don't use OFFSET or INDEX/MATCH when a simple VLOOKUP will do the trick